Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when the facts needed for the equation are true.
3. Explain the energy changes and the limits of your calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

A hypothetical target needs a **3 km** sideways shift over **5×10^{6} s**. Its mass is **7×10^{8} kg**. Estimate Δv and impulse; **describe the calculation you used** and what it treats as true.

Word help: estimate = find an approximate value

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

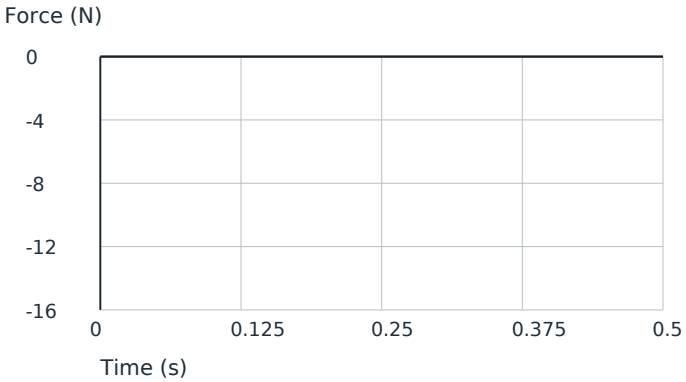
02 Worked example: explain the prediction

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. **Label before**/after diagrams and directions.

Worked reasoning: A **2 kg** cart moving at **3 m/s** stops in **0.5 s**. Momentum change is **-6 kg·m/s**; average force is **-12 N**. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A **1 kg** cart at **4 m/s** right sticks to a **1 kg** cart at rest. **Draw before**/after and **explain which quantity** is conserved.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **2 kg** cart at **+6 m/s** sticks to a **1 kg** cart at **-3 m/s**. **Find final velocity** and kinetic energy transformed. Organize the known quantities on the before-and-after diagram.

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A report says a collision is elastic because total momentum is equal before and after. What extra calculation is needed to **check this conclusion**?

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A cart has momentum **-24 kg·m/s** and mass **4 kg**. **Find velocity** and its new momentum if velocity reverses at unchanged speed.
Use before and after snapshots to organize the momentum account.

Word help: velocity = rate of position change, with direction

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

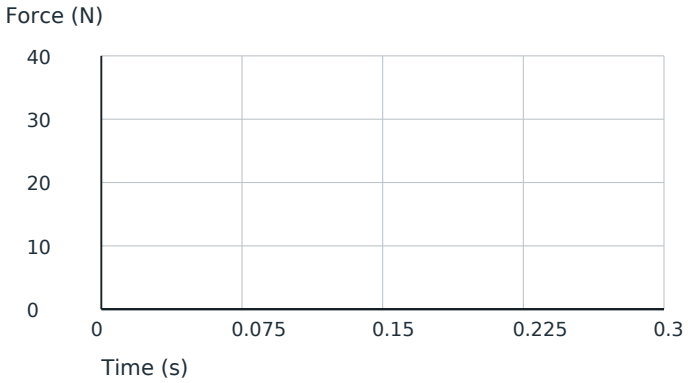
Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A triangular force pulse has peak **40 N** and duration **0.30 s**. It acts on a **2 kg** cart initially at **-1 m/s**. **Find impulse** and final velocity. Organize the known quantities on the before-and-after diagram.

Sentence starter: I found … [unit]. This value means … in this situation.



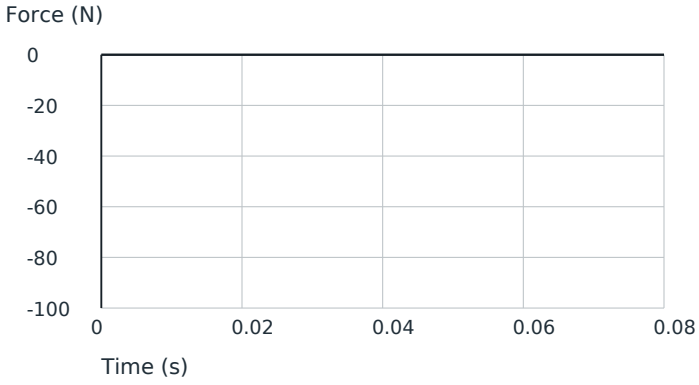
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **0.4 kg** ball rebounds from **+5** to **-3 m/s**. A triangular force pulse lasts **0.08 s**. **Find impulse**, average force and peak-force magnitude. Organize the known quantities on the before-and-after diagram.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3